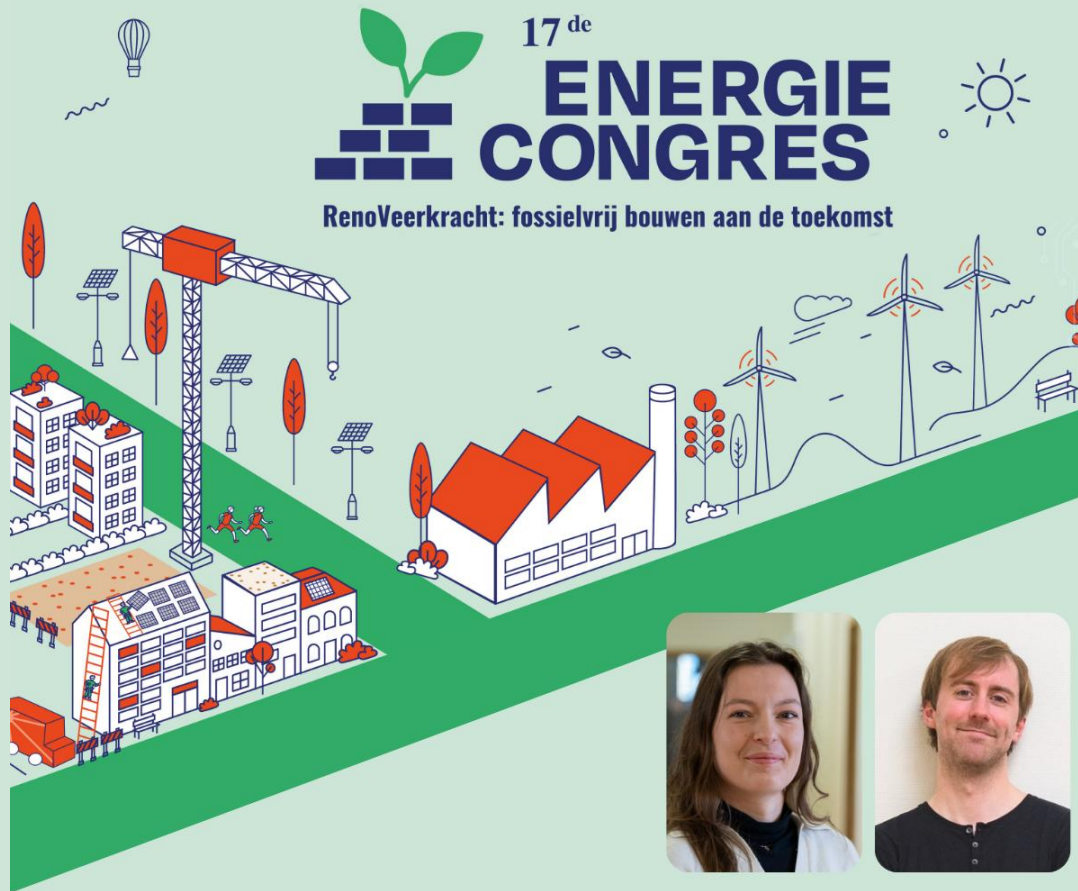




# 17<sup>de</sup> ENERGIE CONGRES

RenoVeerkracht: fossielvrij bouwen aan de toekomst



*RE-ENNOVATE, een gepersonaliseerde en contextbewuste  
AI-tool voor energiebesparende renovatiesuggesties*

*Emma Nuyts, IDLab-UGent-imec & Marc Delghust, Scone.ai*



**Emma Nuyts** is PhD-student bij onderzoeksgroep IDLab (UGent-imec), onder supervisie van Femke Ongenae en Sofie Vanhoecke.

Haar onderzoek gaat over het combineren van semantiek en machinaal leren voor geoptimaliseerd gebouwbeheer.



**Marc Delghust** 'Head of Research' bij Scone & senior post-doc aan UGent - onderzoeksgroep Bouwfysica

Bouwer van R&D trajecten & partnerschappen.

Domeinen: energiestaat van gebouwen, performance gap, building-stock, ...



# RE-ENNOVATE

Re-innovate the process of Energetic Renovation



2022-2025

# AI voor het genereren van renovatiepaden

Machine Learning = data-gedreven analyse

Semantiek = kennis-gedreven analyse

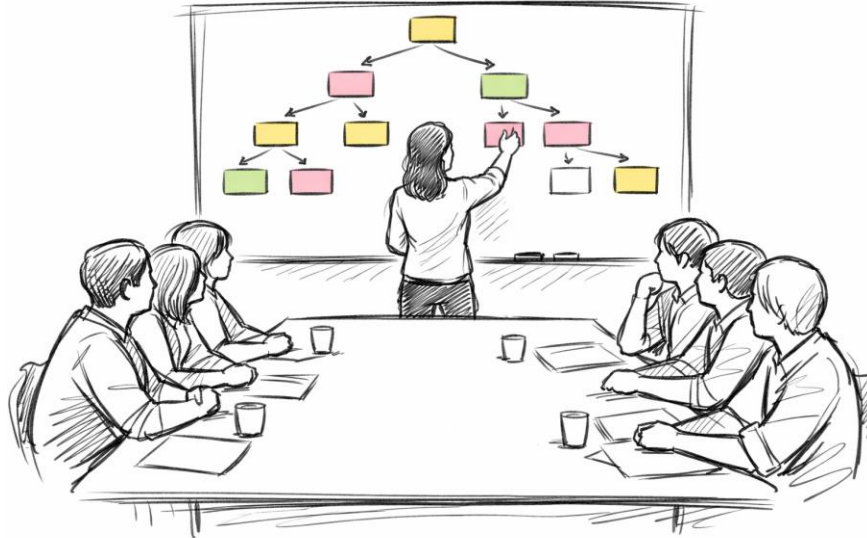
Renovatiesuggesties zijn héél kennis-gedreven: domeinexperten bezitten kennis, is niet (in detail) gedocumenteerd

→ domeinkennis capteren om AI op te baseren

# Kennisverwerving workshops

5 workshops doorheen Vlaanderen met 4-5 participanten per workshop

Variërende achtergronden: architecten, renovatiecoaches, onderzoekers...



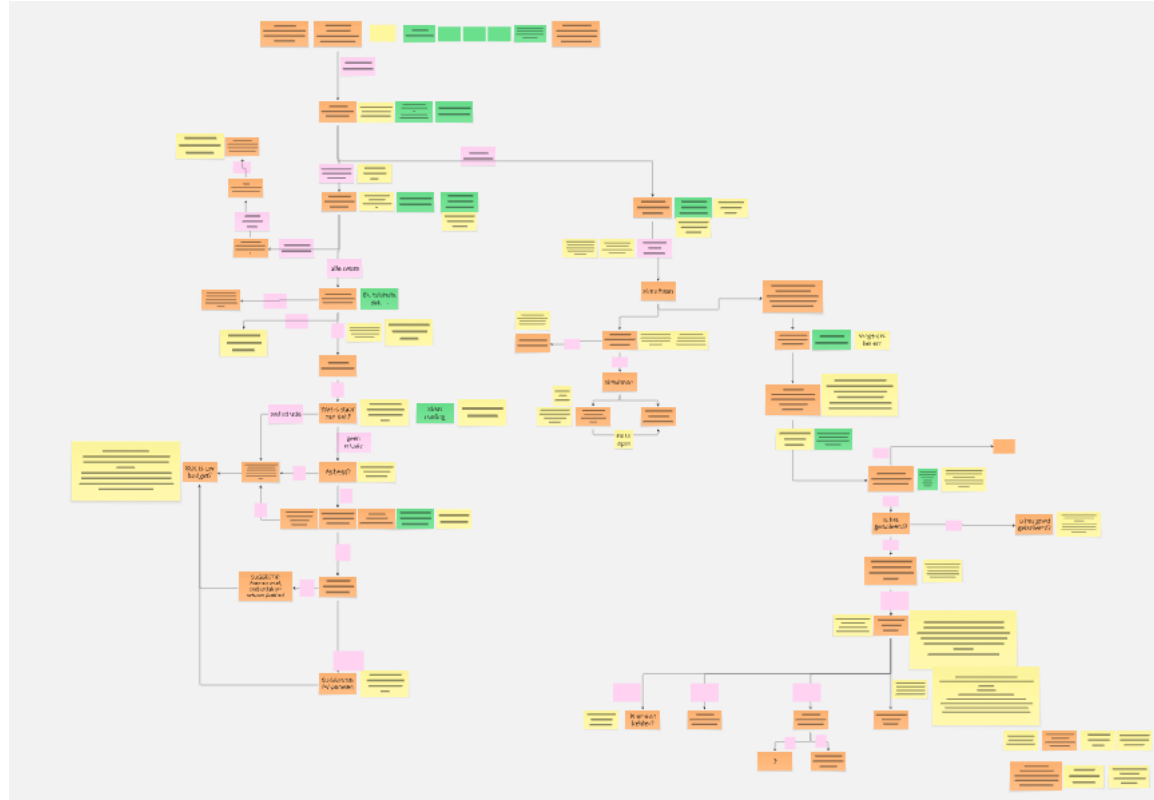
# Kennisverwerving workshops

**Doel: welke informatie is nodig om tot een goed, persoonlijk renovatievoorstel te komen?**

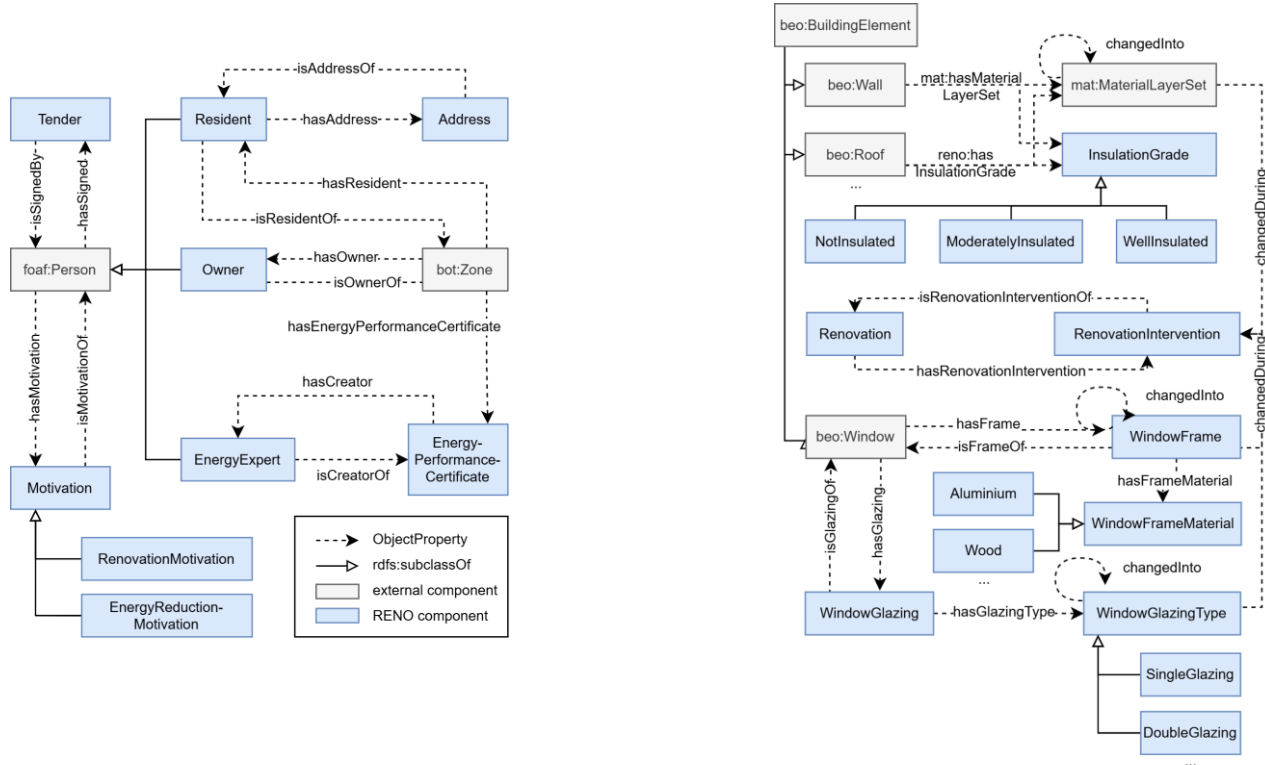
Stel dat je een klant bezoekt die renovatiemaatregelen wil om zijn woning energiezuiniger te maken. Je kent de klant en de woning nog niet, dus je moet alle nodige informatie opvragen over gebouw, gebruik, voorkeuren en budget.

Ga ervan uit dat de klant alle gevraagde gegevens kan aanreiken.

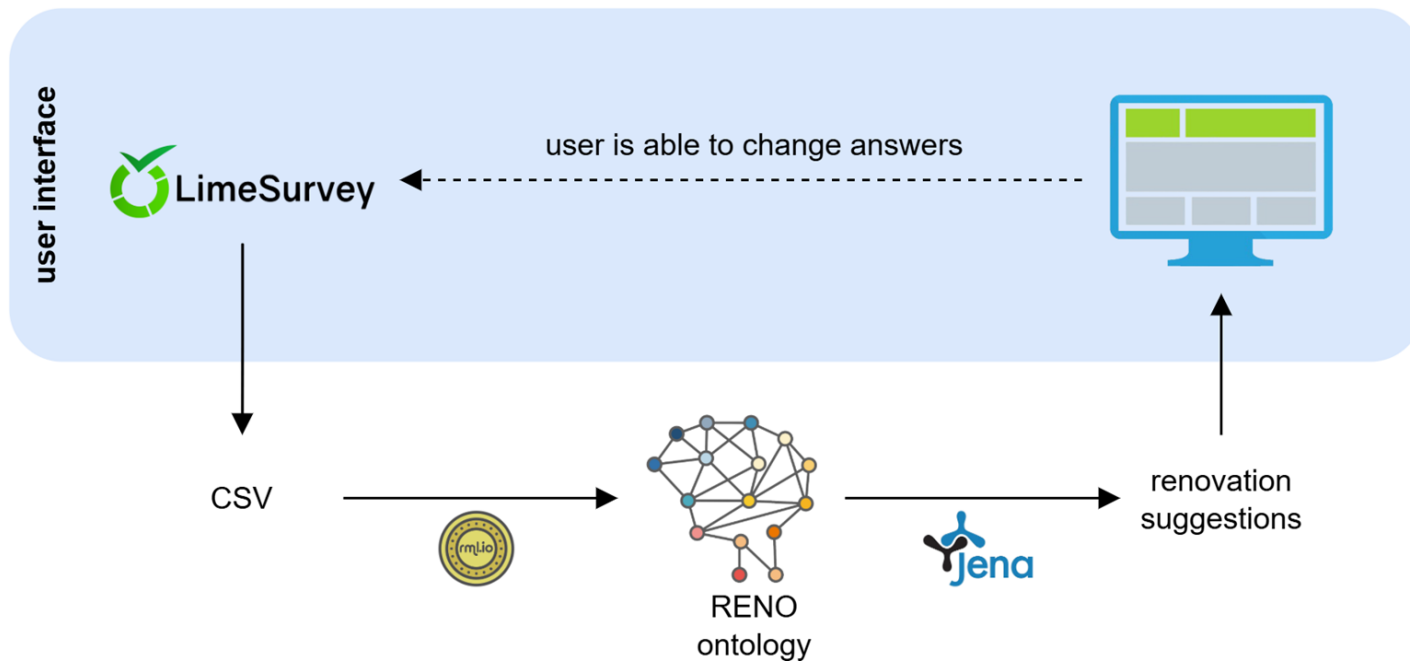
# Kennisverwerking workshops



# Renovatie en ENergie Optimalisatie (RENO) ontologie



# OnToRenovate



# OnToRenovate

\* 7 G01Q07

|                                                                | Yes                   | No                               |
|----------------------------------------------------------------|-----------------------|----------------------------------|
| Is an up-to-date EPB or EPC report available for the property? | <input type="radio"/> | <input checked="" type="radio"/> |

\* 9 G01Q09

|                                            | Yes                              | No                    |
|--------------------------------------------|----------------------------------|-----------------------|
| Do you know the EPC label of the property? | <input checked="" type="radio"/> | <input type="radio"/> |

\* 10 G01Q10 Input EPC label

🔗 Choose one of the following answers

A ▼

\* 11 G01Q11a Which year the property was built in?

🔗 Please check the format of your answer.

1910

# OnToRenovate

## Renovation suggestions

By carrying out your renovations in the right order, you will get the most out of your investment and increase your living comfort without unnecessary costs. The best first step is to insulate your home. A well-insulated exterior shell will reduce your energy consumption significantly, which can lead to serious annual savings. This means a warmer home in winter and a cooler home in summer, without major temperature fluctuations. So you lay a solid foundation for further energy-saving measures. If your home is well insulated, it is time to take sustainable techniques a step further. Solar panels can significantly reduce your electricity bill, while a heat pump lowers your heating costs. Plus, there are grants and subsidies available, making this investment pay off faster than you initially think.

The prices shown are estimates based on the entered building characteristics. These prices include materials, installation, waste disposal and any demolition, but exclude VAT. Premiums are not included in this estimate. For information about premiums, please visit [www.premiezoeeker.be](http://www.premiezoeeker.be). For more detailed renovation advice, please contact an 'Energiehuis' via [www.mijnenergiehuis.be](http://www.mijnenergiehuis.be).

Your renovation budget: € 20.000

✓ Advised ⓘ

? Advised with caveat ⓘ

✗ Over budget ⓘ

? STEP 1 place roof underlayment

€ 16.500

Additional information -

### Alternative 1

### Alternative 2

✓ STEP 2 insulate wall on interior

Variant ⓘ

€ 3.500

Additional information -

✗ STEP 2 insulate wall on exterior

Variant ⓘ

€ 4.700

Additional information -

✗ STEP 3 place emission system on low temperature € 18.000

Additional information -

✗ STEP 3 place emission system on low temperature € 18.000

Additional information -

✗ STEP 4 place heatpumpboiler

€ 4.000

Additional information -

✗ STEP 4 place heatpumpboiler

€ 4.000

Additional information -

Total cost: € 42.000

Total cost: € 43.200

Last update of OnToRenovate: 28/11/2025

# OnToRenovate

## Renovation suggestions

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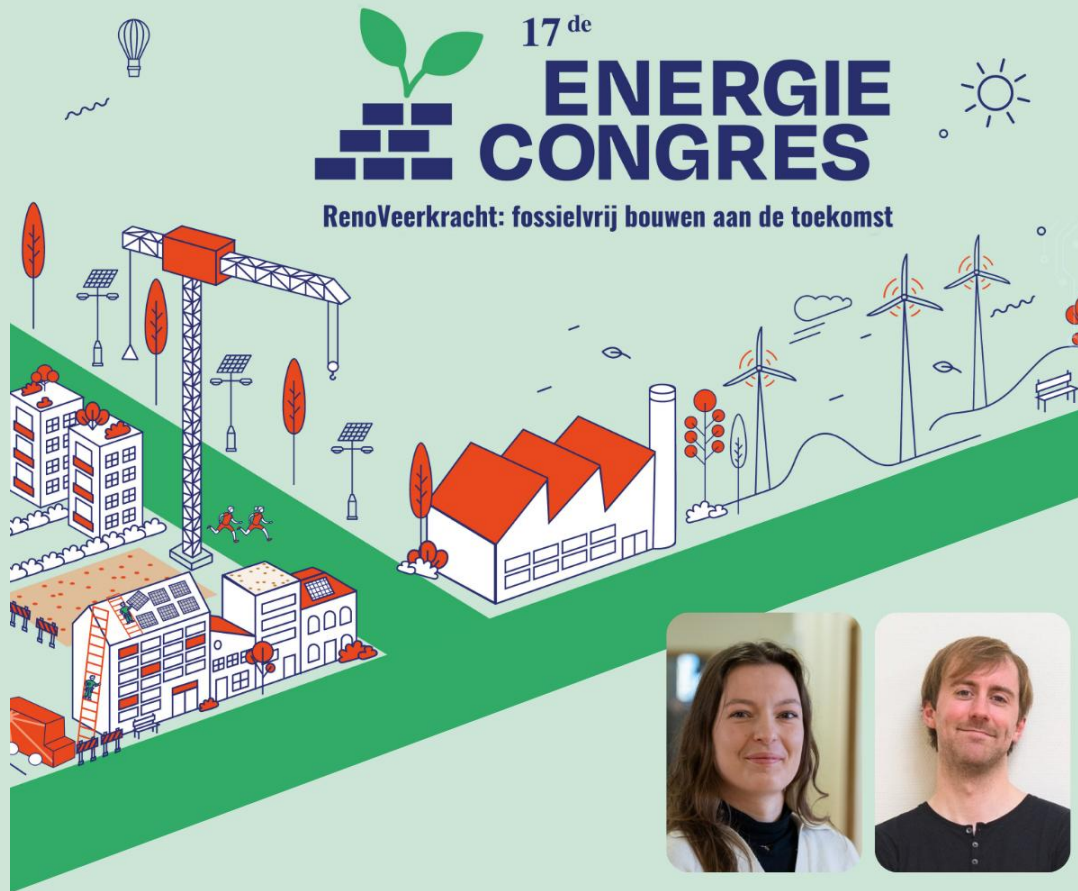
Last update of OnToRenovate: 28/11/2025



# Gebruikerstesten

Twee groepen voor testfase:

- met renovatiecoaches: zijn de gegenereerde suggesties correct?  
workshops met 5 renovatiecoaches
- met gebouweigenaren: zijn alle vragen duidelijk?  
test met 22 gebouweigenaren



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